Model Specification Continued

POLS 602
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Mike Burnham
Texas A&M University

Review

1. What does it mean for a parameter to be identified?
2. What does a fixed effect absorb, and what does it not?

Today

- How do we model relationships that are curved instead of straight?
- How do we model effects that depend on another variable?
- What if we do not want to choose a functional form at all?
- What if we care about more than the average outcome?

The Log-Linear Model

Skewed Outcomes

- Income, wages, prices, and populations are right-skewed
- Effects are often proportional instead of additive
- *Does one more year of education add a fixed dollar amount, or a fixed percentage?*

Logarithm Refresher

- $\ln(y)$ is the power we raise $e \approx 2.718$ to in order to get y, so $e^{\ln y} = y$
- Logs turn products into sums: $\ln(ab) = \ln a + \ln b$
- Logs turn powers into products: $\ln(a^b) = b \ln a$
- For small r, $\ln(1 + r) \approx r$

Ln(y) Models Proportional Effects

Suppose the effect of x on y is a fixed rate, rather than amount (e.g. compounding interest):

$$\ln(y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Exponentiate both sides:

$$y_i = e^{\beta_0}\, e^{\beta_1 x_i}\, e^{\varepsilon_i}$$

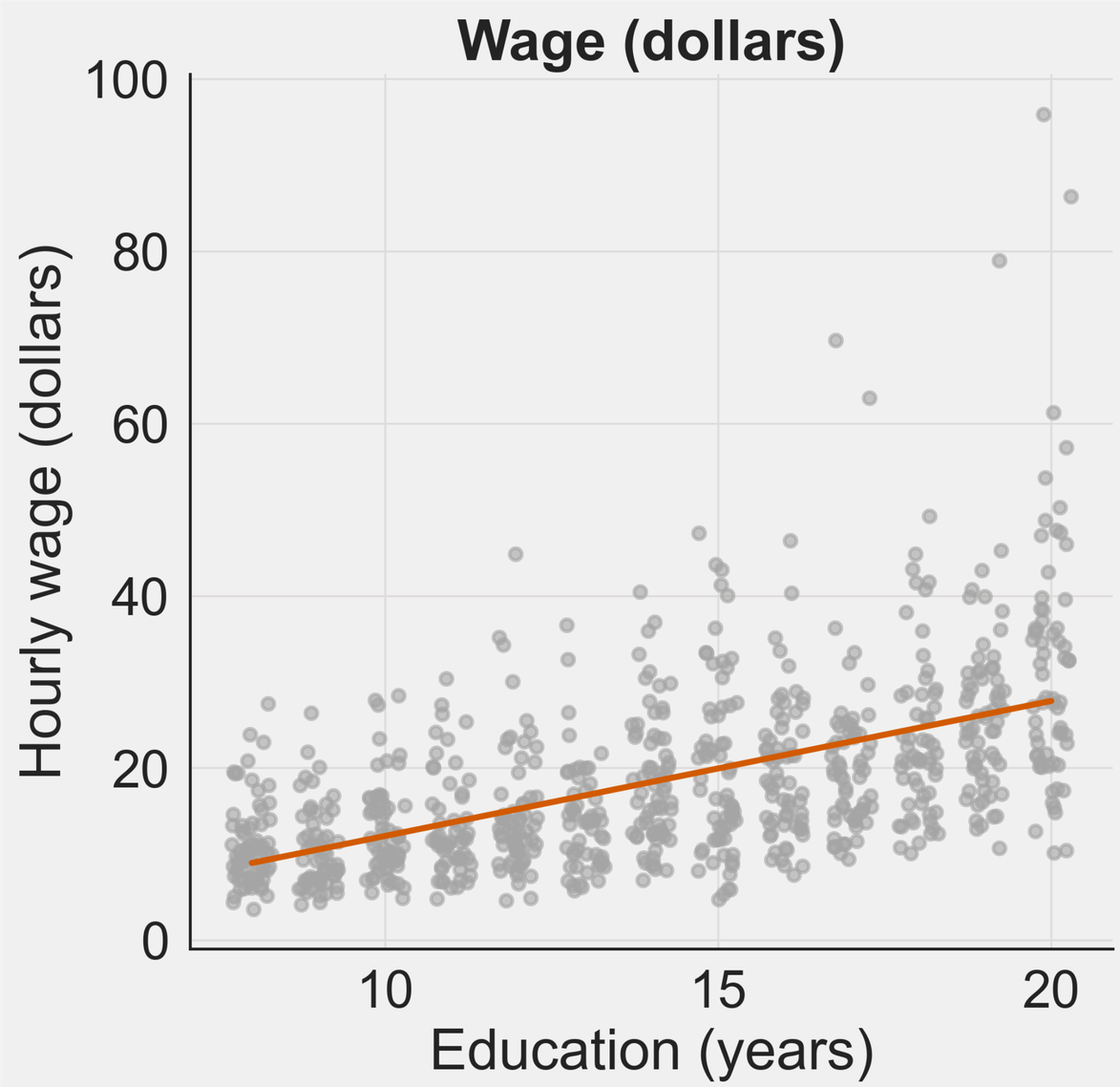
Raise x_i by one unit. y_i is multiplied by e^{β_1}.

The Log-Linear Model

$$\ln(y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- A one-unit increase in x changes y by about $100 \times \beta_1$ percent
- Exactly: $100 \times (e^{\beta_1} - 1)$ percent

Wages in Levels and Logs





Four Log Specifications

Model	Equation	Interpretation of β_1
Linear-linear	$y = \beta_0 + \beta_1 x$	One unit of x changes y by β_1
Log-linear	$\ln y = \beta_0 + \beta_1 x$	One unit of x changes y by $100\beta_1$ percent
Linear-log	$y = \beta_0 + \beta_1 \ln x$	One percent of x changes y by $\beta_1/100$

Quadratic Regression

Effects That Change Direction

- Wages rise with experience, then level off, and may fall
- A line cannot do this
- *What happens to a worker's wage in her first year versus her thirtieth?*

The Quadratic Model

$$y_i = \beta_0 + \beta_1 \text{exper}_i + \beta_2 \text{exper}_i^2 + \varepsilon_i$$

- Add the squared term as one more column of $\mathbf{X}$
- $\beta_2 < 0$ gives an inverted U
- $\beta_2 > 0$ gives a U

What Is a Marginal Effect?

- **Marginal effect**: how much the predicted y changes when x rises by one unit, other variables held fixed
- In a linear model, it is the slope β, the same everywhere
- In a curved model, it is the slope of the curve at a particular value of x
- Slopes of curves come from derivatives

Marginal Effect of Experience

Differentiate the quadratic model with respect to experience:

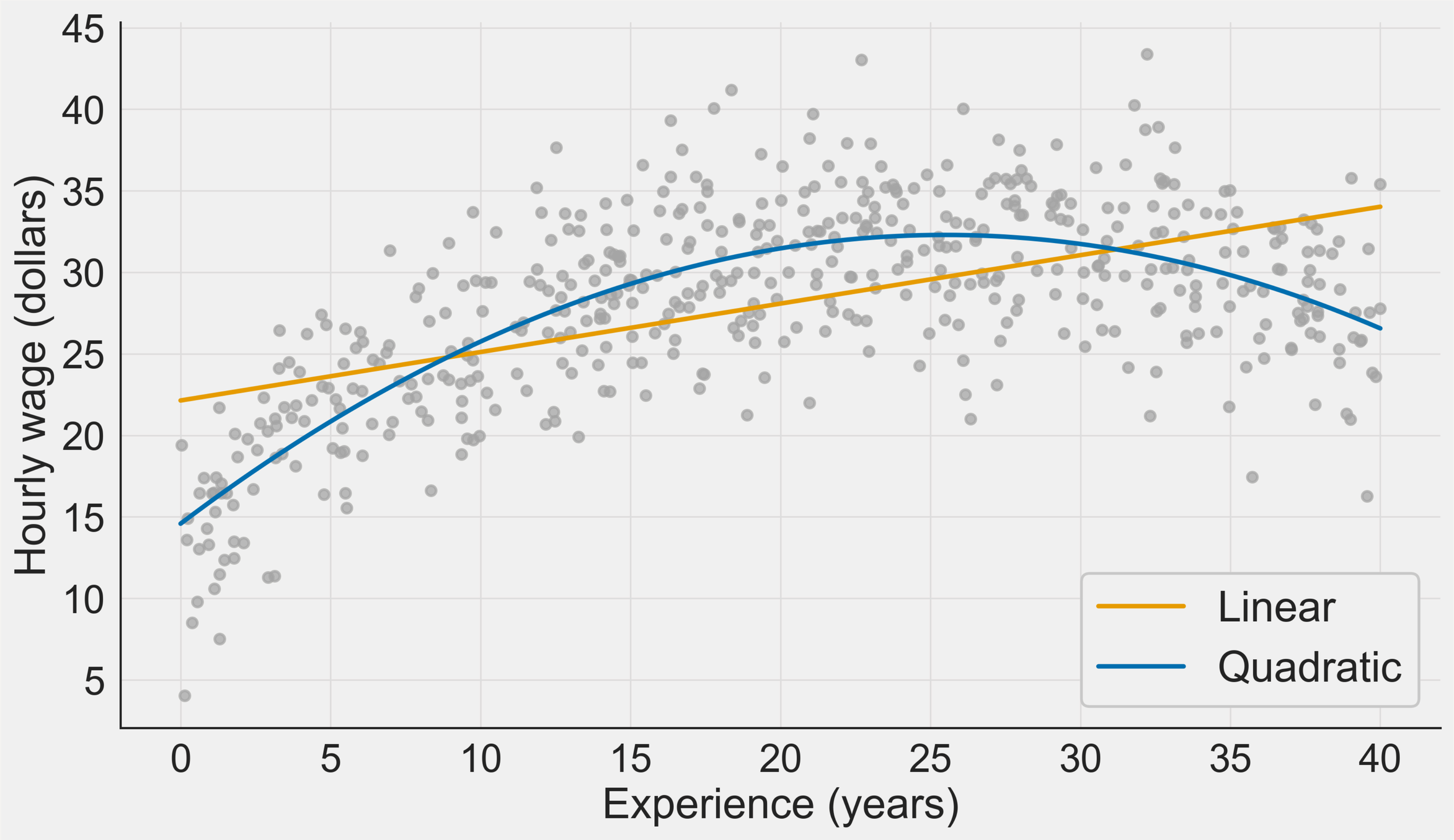
$$\frac{\partial y}{\partial \text{exper}} = \beta_1 + 2\beta_2 \, \text{exper}$$

The effect of one more year depends on how many years you already have.

The turning point is where the marginal effect is zero:

$$0 = \beta_1 + 2\beta_2 \, \text{exper}^* \qquad \Rightarrow \qquad \text{exper}^* = -\frac{\beta_1}{2\beta_2}$$

Wages and Experience



Interactions and Marginal Effects

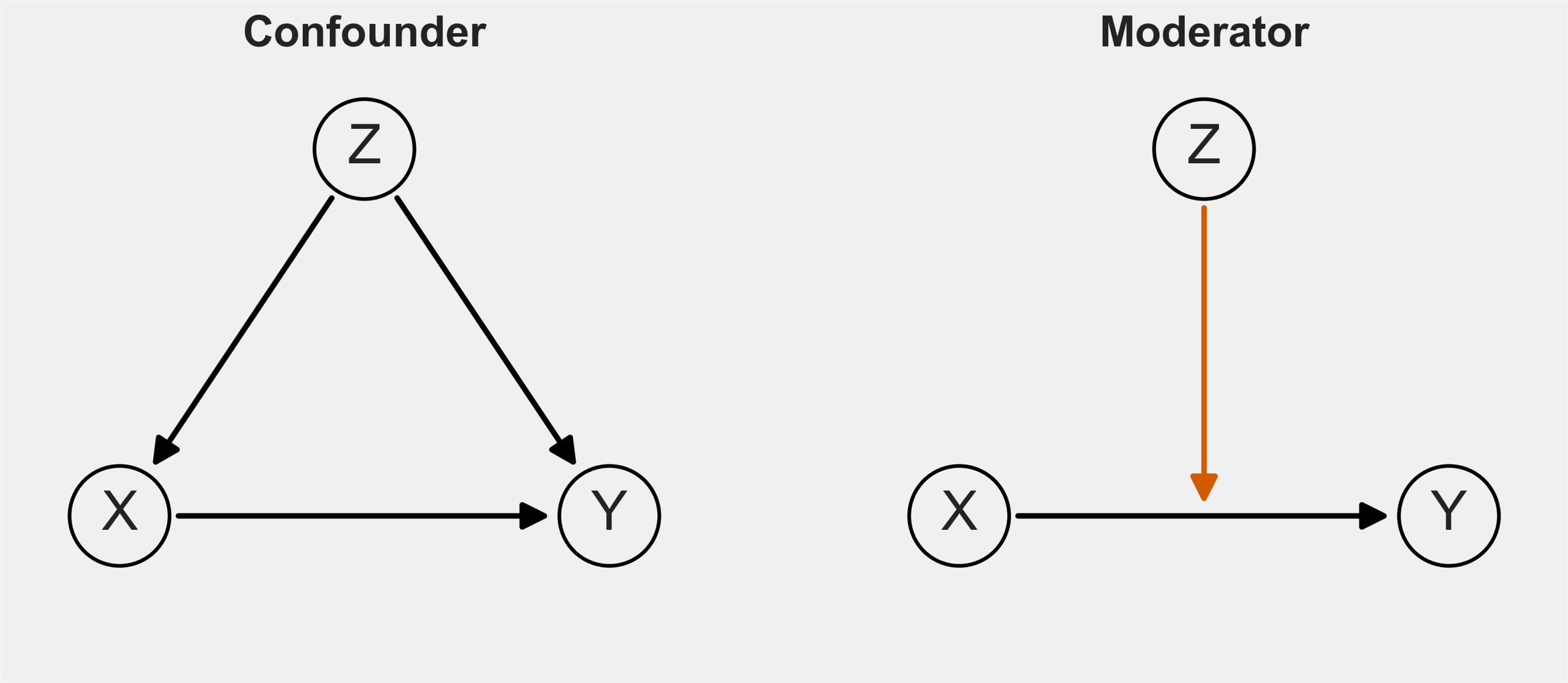
Moderators

- **Moderator**: a variable that changes the effect of X on Y
- The effect of a policy may differ in rich and poor states
- The effect of an ad may differ for strong and weak partisans

Example: Party Cues

- Respondents read a policy proposal
- Half are randomly told that the Democratic Party supports it
- The outcome is support for the proposal
- *Does the cue have the same effect on Democrats and Republicans?*

Moderators vs. Confounders



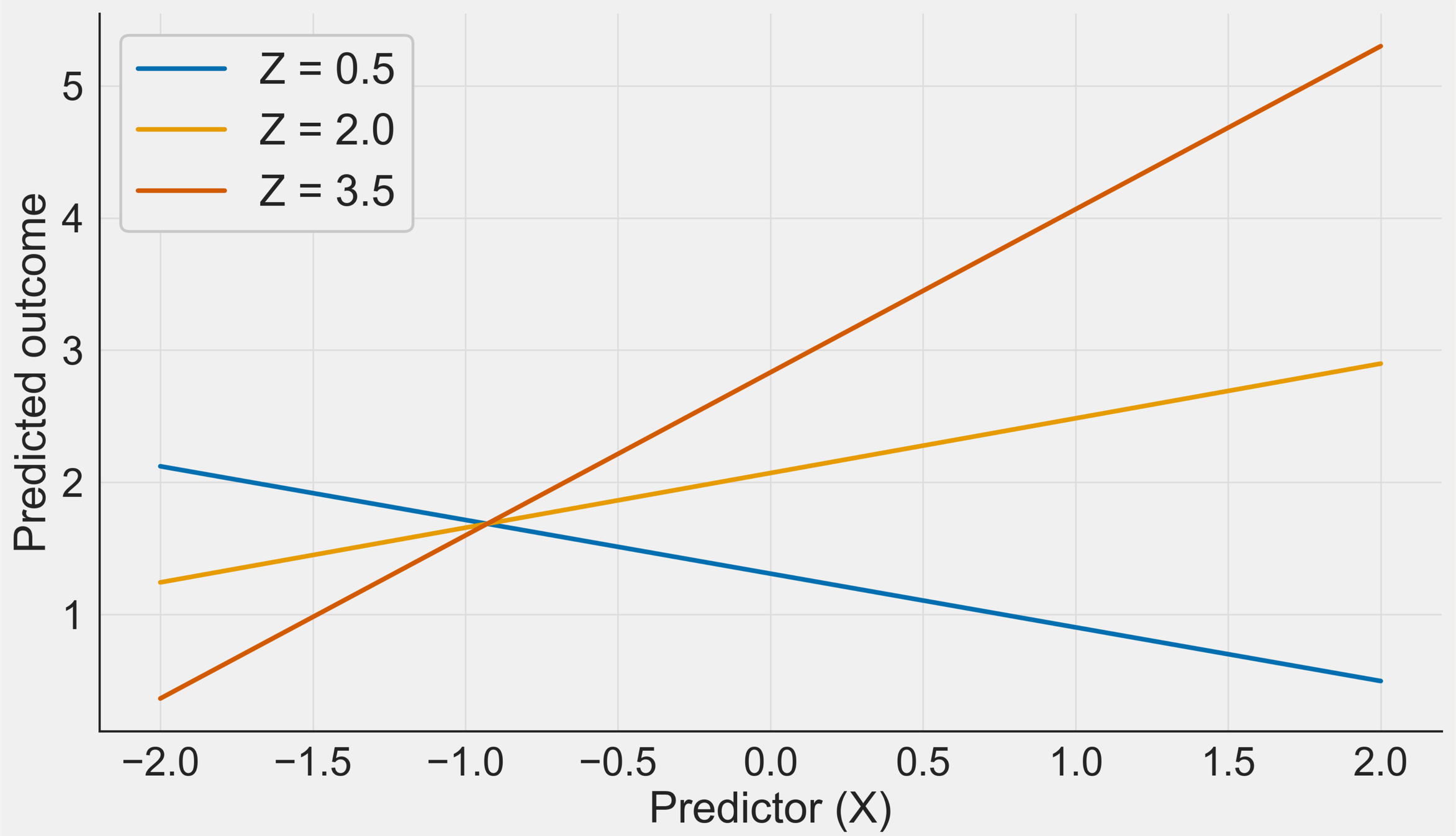
The Interaction Model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \varepsilon_i$$

$$\frac{\partial y}{\partial x} = \beta_1 + \beta_3 z$$

- β_1 is the effect of X when $Z = 0$
- β_3 is how much that effect changes per unit of Z

Predicted Values by Moderator



The Marginal Effect Is an Estimate

- $\hat{\beta}_1 + \hat{\beta}_3 z$ is computed from the data, so it would differ in another sample
- Like any estimate, it needs a standard error and a confidence interval
- The table reports standard errors for $\hat{\beta}_1$ and $\hat{\beta}_3$ separately, not for their combination
- *How do we find the standard error of a sum of two estimates?*

The Variance of a Sum of Estimates

From a previous lecture: $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$. For a sum, in general:

$$\mathrm{Var}(A + zB) = \mathrm{Var}(A) + z^2\,\mathrm{Var}(B) + 2z\,\mathrm{Cov}(A, B)$$

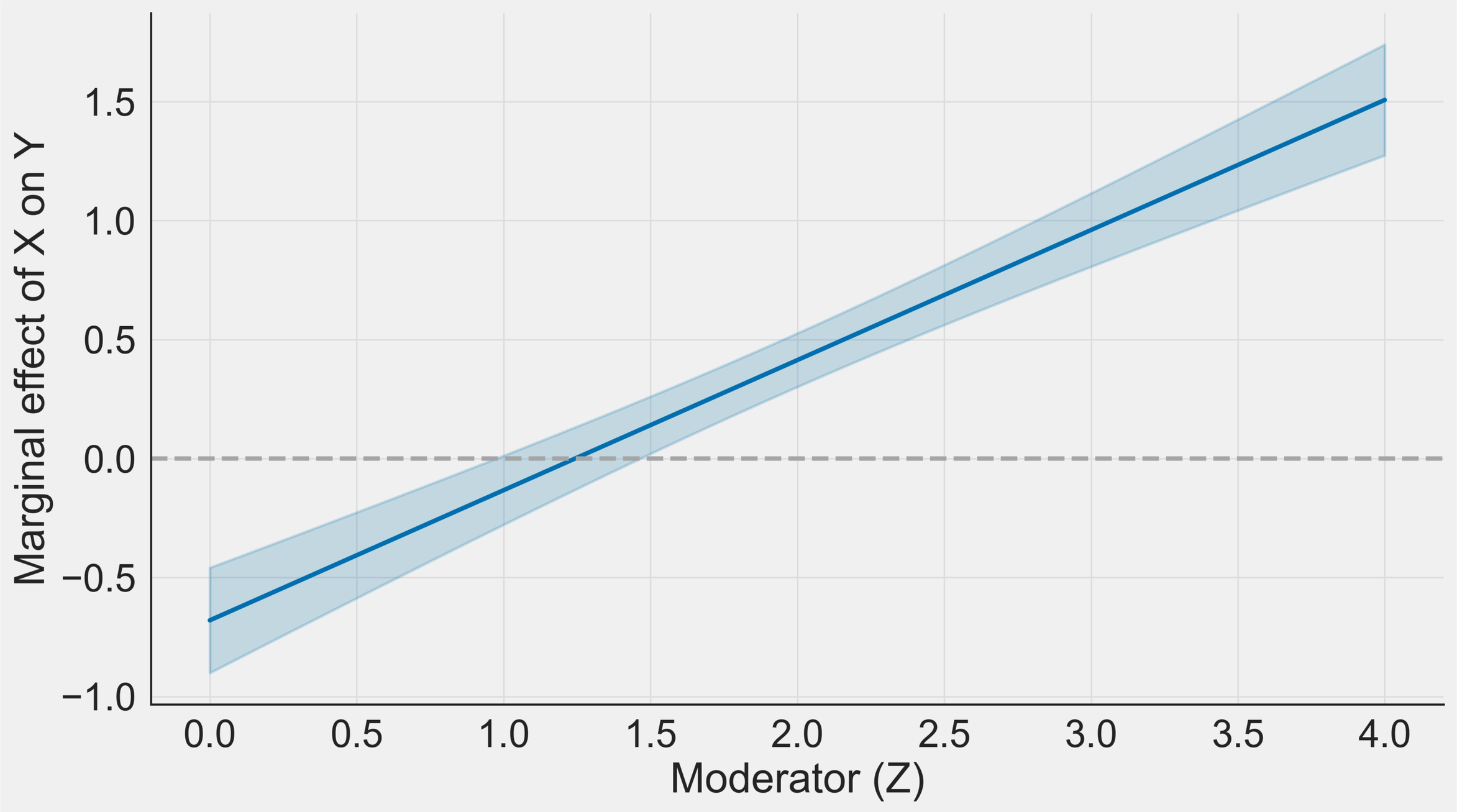
- A is $\widehat{\beta}_1$, B is $\widehat{\beta}_3$, and z is a constant we choose
- The covariance term appears because both come from the same regression

Uncertainty in Marginal Effects

$$\widehat{\text{Var}}\left(\hat{\beta}_1 + \hat{\beta}_3 z\right) = \widehat{\text{Var}}(\hat{\beta}_1) + z^2 \widehat{\text{Var}}(\hat{\beta}_3) + 2z \widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_3)$$

- The standard error is the square root, and the 95% interval is the estimate ± about two standard errors
- The standard error changes with z
- The regression table alone does not give it to you

Effects Change With Z



Interactions With Dummy Variables

- Dummy × continuous: separate slopes by group
- Dummy × dummy: the effect of one group depends on the other
- The difference-in-differences model is a dummy × dummy interaction

Heterogeneous Treatment Effects

- **Average treatment effect**: the mean effect of the treatment across all units
- **Heterogeneous treatment effects**: effects that differ across units
- **Conditional average treatment effect**: the mean effect among units with a given value of Z
- A moderator is a variable along which treatment effects are heterogeneous

Interactions Estimate CATEs

Let D_i be the treatment and Z_i the moderator:

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 Z_i + \beta_3 D_i Z_i + \varepsilon_i$$

$$\text{CATE}(z) = \beta_1 + \beta_3 z$$

- β_3 is the difference in the treatment effect between units one unit apart on Z

Party Cues: Reading the Model

Let D_i be the cue and $Z_i = 1$ for Democrats, 0 for Republicans:

Quantity	Meaning
β_1	Effect of the cue among Republicans
$\beta_1 + \beta_3$	Effect of the cue among Democrats
β_3	Difference in the effect between parties
$\beta_1 + \beta_3 \bar{z}$	Average effect in the sample

What the Interaction Identifies

- Randomizing D identifies the effect of D at each value of Z
- Z must be measured before treatment
- Z was not randomized, so β_3 is not the causal effect of Z
- Interactions need more data than average effects
- Testing many moderators produces false positives

LOESS

Choosing the Shape

- Logs and quadratics require us to name the shape in advance
- *What if the relationship rises, plateaus, and then falls?*
- A nonparametric method lets the data determine the shape

Parametric vs. Nonparametric

$$y_i = f(x_i) + \varepsilon_i$$

- Parametric: assume f has a form, such as $\beta_0 + \beta_1 x + \beta_2 x^2$
- Nonparametric: estimate f without assuming a form

Local Regression

- Fit a regression at a single value of x, using only the data near it
- Repeat at many values of x
- Connect the fitted values

Neighborhoods and Weights

For a target value x_0:

1. Choose the share of the data closest to x_0. This share is the span.
2. Give each neighbor a weight that falls with its distance from x_0.

$$w_i = \left(1 - \left(\frac{|x_i - x_0|}{h}\right)^3\right)^3 \quad \text{for } |x_i - x_0| < h$$

h is the distance to the farthest neighbor. Weights outside it are zero.

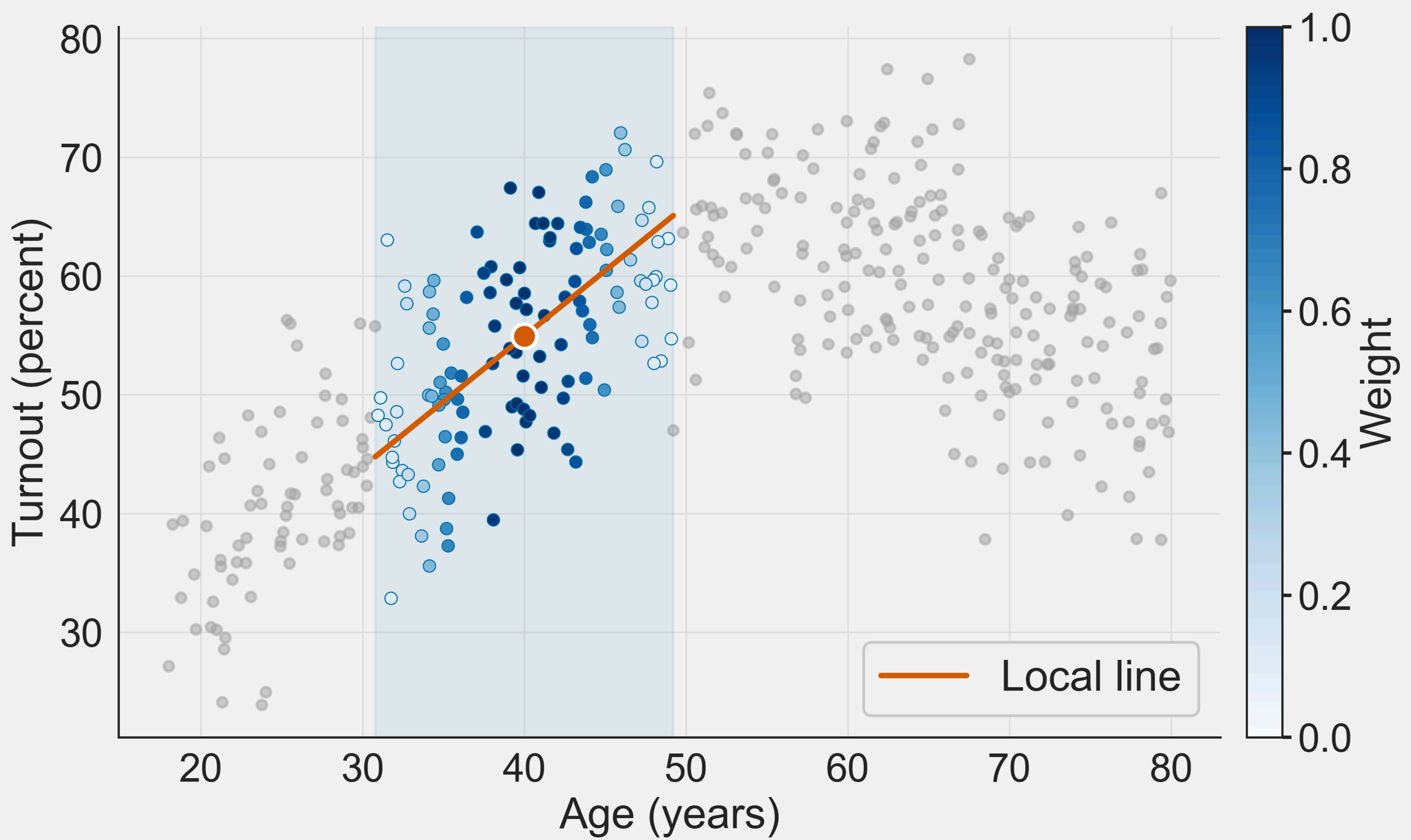
The Local Fit

At each x_0, minimize a weighted sum of squared errors:

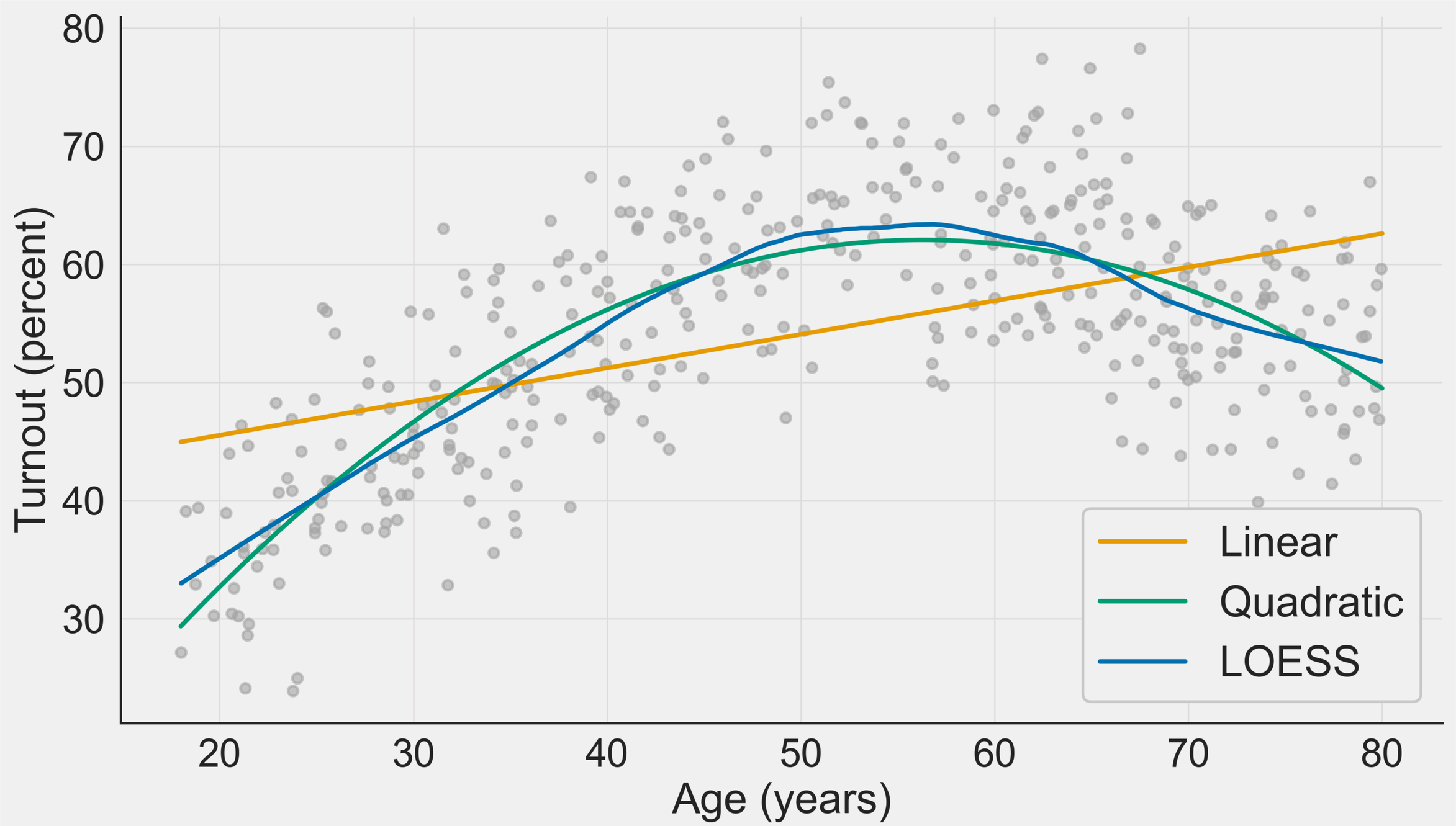
$$\sum_{i=1}^{n} w_i (y_i - a - b\, x_i)^2$$

The fitted value at x_0 is $\widehat{f}\,(x_0) = \hat{a} + \hat{b}\, x_0$.

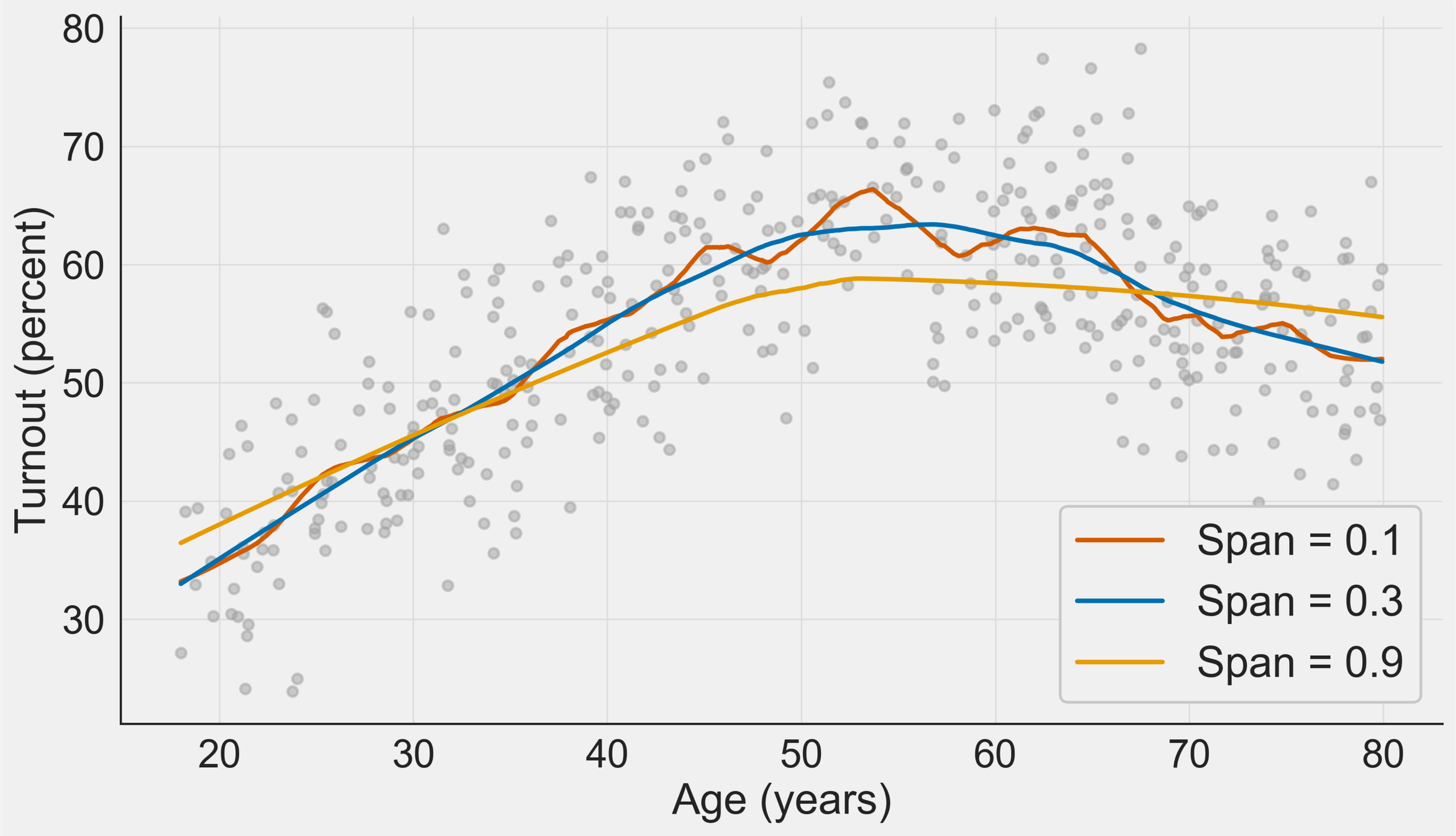
One Local Fit



LOESS Finds the Shape



The Span Controls Smoothness



What LOESS Gives Up

- No coefficients to report. The output is a curve
- The span is a choice that changes the answer
- Estimates are unreliable near the edges of the data
- It handles one or two predictors well, but not many
- Extrapolating beyond the data is not possible

Quantile Regression

Beyond the Mean

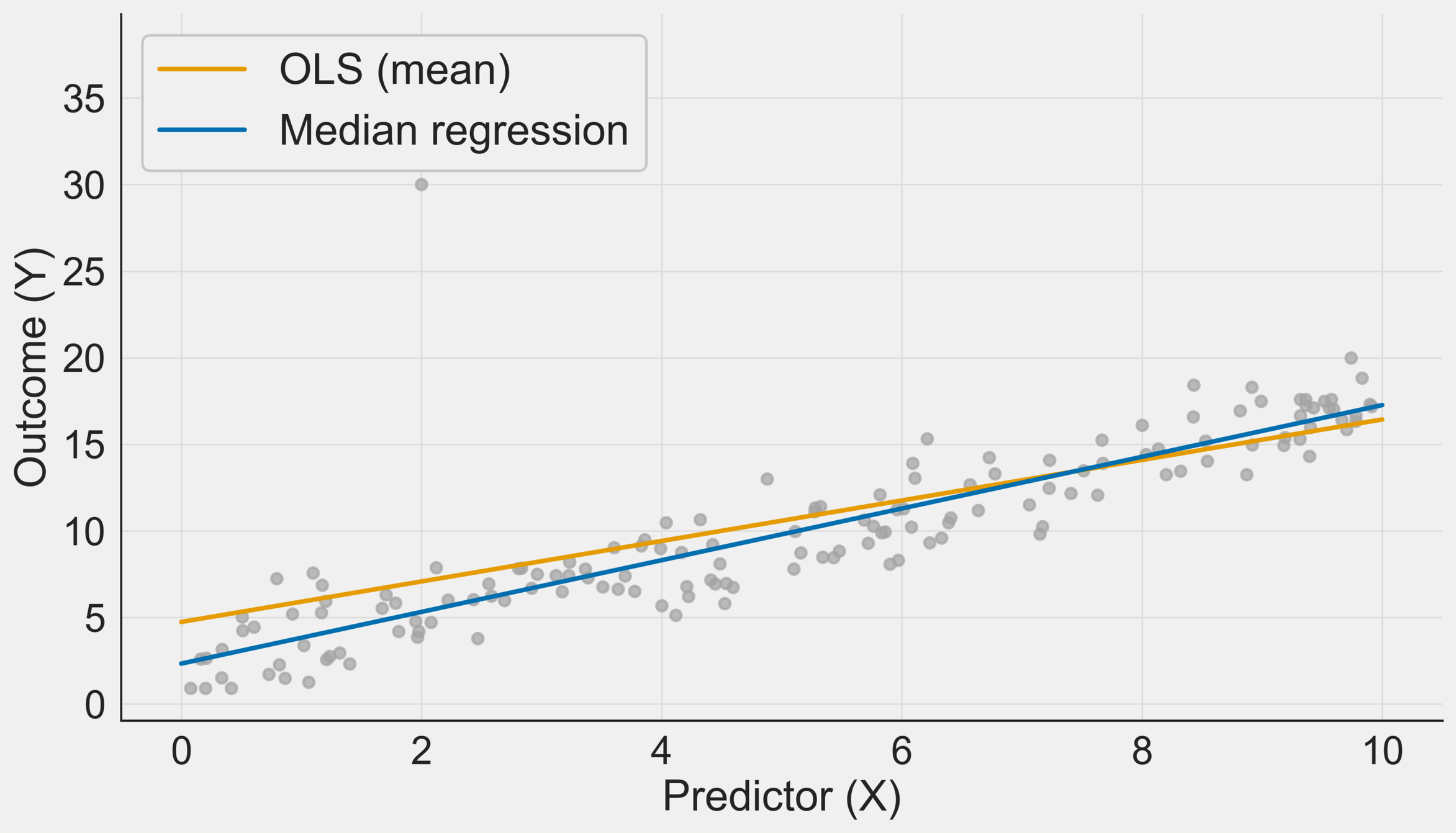
- OLS estimates the conditional mean of y
- *Does education raise the wages of low earners as much as high earners?*
- The mean can hide effects that differ across the distribution

The Median Instead of the Mean

- The mean is the value that minimizes the sum of squared deviations
- The median is the value that minimizes the sum of absolute deviations

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Median Regression Resists Outliers



The Quantile Loss Function

For quantile τ between 0 and 1, residuals are weighted asymmetrically:

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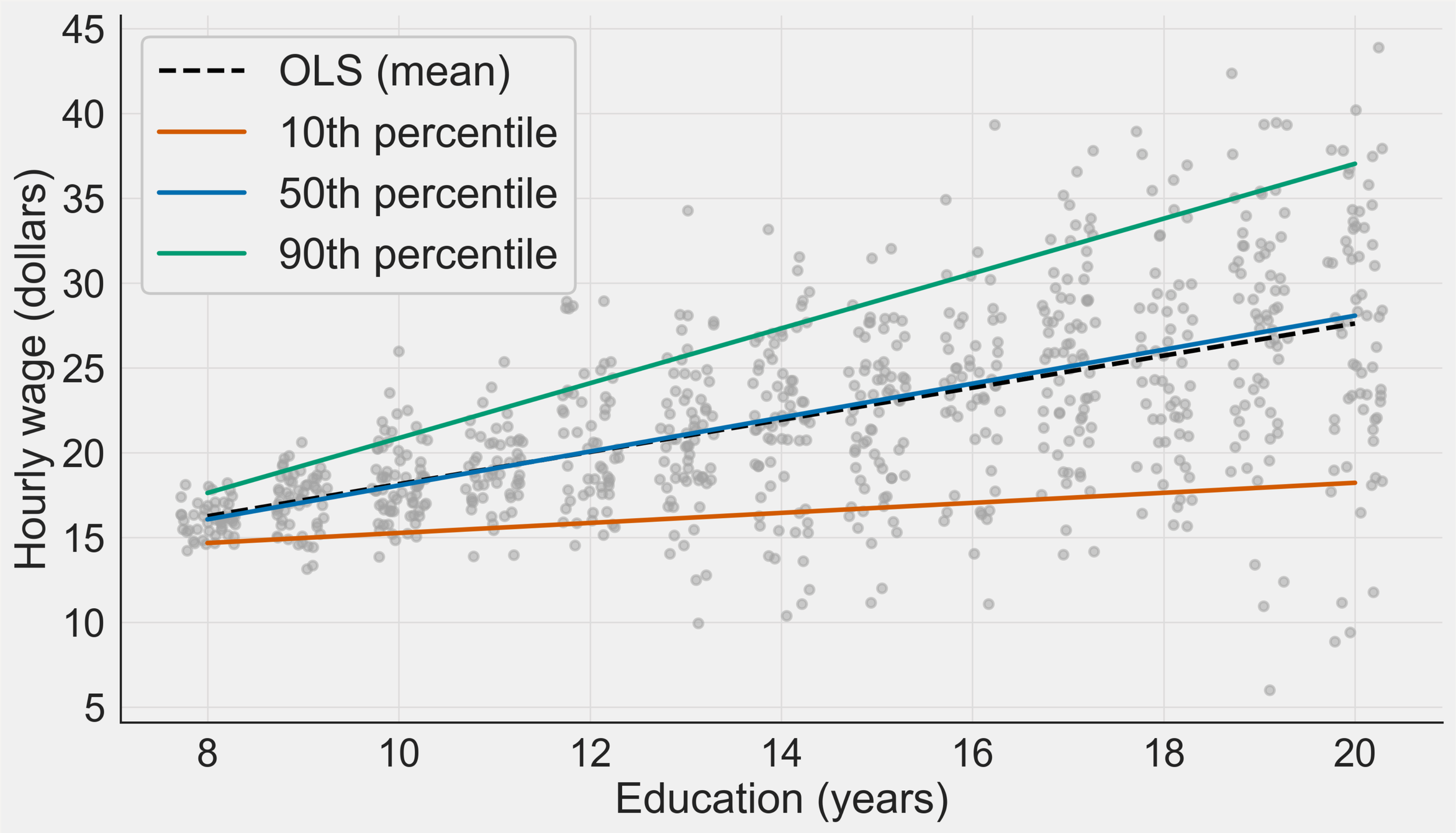
Positive residuals get weight τ. Negative residuals get weight $1 - \tau$.

The Quantile Regression Model

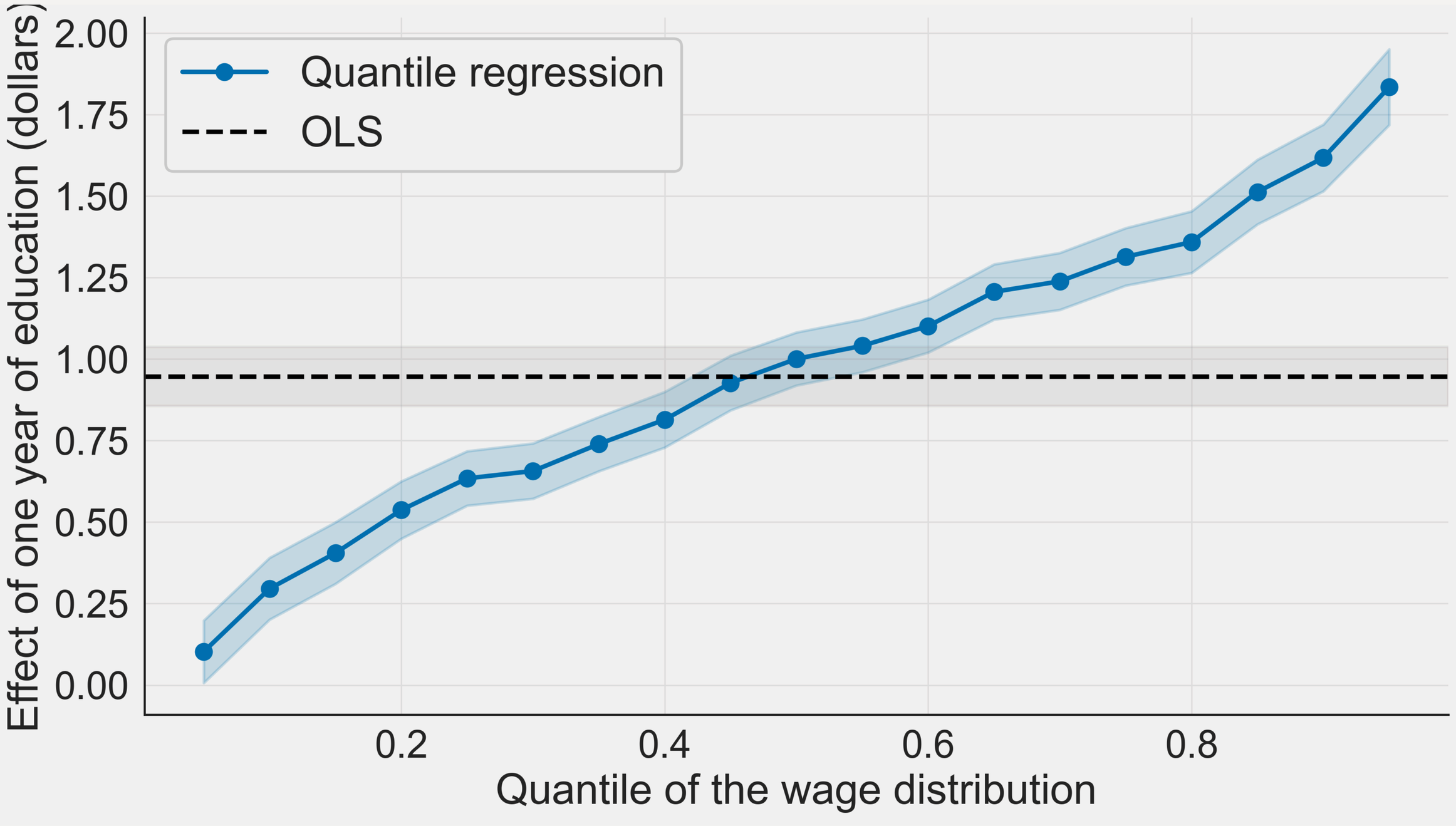
$$Q_\tau(y_i \mid \mathbf{x}_i) = \beta_0(\tau) + \beta_1(\tau)\, x_i$$

- Q_τ is the τth conditional quantile of y
- Each quantile has its own coefficients
- If the spread of y is the same at every x, the slopes are all the same

Wages Across the Distribution



Coefficients by Quantile



Interpreting Quantile Coefficients

- $\beta(\tau)$ describes a quantile of the outcome, not a type of person
- A person at the 10th percentile before a treatment is not necessarily at the 10th percentile after
- Quantile lines can cross, which should not happen
- A causal reading still needs the identification assumptions

Summary

- Logs and quadratics model curves and stay linear in the parameters, with coefficients read as percent changes and marginal effects
- Interactions let the effect of one variable depend on another, which is how we estimate heterogeneous treatment effects, and the uncertainty of the marginal effect depends on the covariance of the coefficients
- LOESS estimates the conditional mean with many weighted local regressions, so the data determine the shape, at the cost of having no coefficients
- Quantile regression models a chosen quantile of the outcome, which shows effects that differ across the distribution and resists outliers at the median

Key Definitions

Key Definitions

- **Log-linear model**: $\ln y$ on x, with β_1 read as a percent change in y
- **Elasticity**: the percent change in y for a one percent change in x, the coefficient in a log-log model
- **Quadratic regression**: a model with x and x^2, allowing a curve with one turning point
- **Marginal effect**: the change in predicted y from a one-unit change in x, which may depend on other variables
- **Turning point**: the value of x where a quadratic changes direction

Key Definitions

- **Moderator**: a variable that changes the effect of X on Y
- **Interaction**: a product term that lets the effect of one variable depend on another
- **Heterogeneous treatment effects**: treatment effects that differ across units
- **Conditional average treatment effect**: the mean treatment effect among units with a given value of a moderator
- **Parametric model**: a model whose shape is set by a fixed number of parameters

Key Definitions

- **Nonparametric model**: a model that estimates the shape of the relationship without assuming a form
- **LOESS**: a curve built from weighted local regressions
- **Span**: the share of the data used in each local fit
- **Tricube weights**: weights that fall smoothly with distance, reaching zero at the edge of the window
- **Median regression**: regression that minimizes the sum of absolute residuals, estimating the conditional median

Key Definitions

- **Quantile regression**: regression that estimates a chosen conditional quantile of the outcome
- **Check function**: the asymmetric loss that defines the τth quantile